

KEY LAKE, SK, Canada

WMO: 710330

Lat: 57.256N

Lon: 105.618W

Elev: 514

StdP: 95.30

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.4	-39.0	-45.0	0.0	-41.9	-43.0	0.1	-38.6	8.5	-15.3	7.6	-16.7	0.7	270	0.737

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	27.1	16.2	25.3	15.3	23.6	14.6	17.7	24.1	16.7	22.5	15.8	21.3	4.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	11.8	19.0	14.6	11.1	18.1	13.6	10.4	17.5	51.9	24.4	48.9	22.5	46.2	21.3	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.7	-46.5	30.4	2.4	1.8	-48.3	31.7	-49.7	32.8	-51.1	33.8	-52.8	35.1		
(5)			-45.8	19.4	2.2	1.0	-47.4	20.1	-48.7	20.7	-49.9	21.3	-51.5	22.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.5	-22.0	-19.9	-13.4	-2.9	5.9	13.1	16.1	14.2	8.2	0.2	-11.1	-19.3		
	DBStd	15.20	9.26	8.61	8.98	7.57	5.53	4.06	3.17	3.72	4.59	5.04	7.43	9.34		
	HDD10.0	5051	992	836	725	387	149	17	1	7	88	307	634	908		
	HDD18.3	7626	1251	1069	983	636	386	162	83	137	306	564	884	1166		
	CDD10.0	494	0	0	0	2	22	109	191	137	33	1	0	0		
	CDD18.3	27	0	0	0	0	1	4	14	7	1	0	0	0		
	CDH23.3	422	0	0	0	0	21	86	200	99	16	0	0	0		
	CDH26.7	67	0	0	0	0	4	13	31	16	3	0	0	0		
(14) Wind	WSAvg	3.1	2.6	2.8	3.0	3.2	3.4	3.3	3.1	3.1	3.3	3.3	2.9	2.5		
(15) Precipitation	PrecAvg	493	26	24	27	20	33	55	91	74	60	35	31	19		
	PrecMax	606	65	87	59	44	76	131	165	127	134	70	69	45		
	PrecMin	405	14	5	6	0	7	17	28	15	18	16	11	0		
	PrecStd	54	11	16	15	10	18	27	33	25	28	14	14	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	2.0	8.5	18.5	26.0	28.2	29.6	28.6	25.4	16.6	5.5	0.3		
		MCWB	0.1	-0.1	3.0	9.4	13.1	16.0	17.4	17.6	15.3	10.5	2.7	-1.0		
	2%	DB	-2.5	-1.1	5.2	14.1	22.1	25.7	27.3	26.0	21.1	12.2	2.6	-2.0		
		MCWB	-3.1	-2.5	1.1	6.7	11.4	14.8	16.5	16.4	13.6	7.4	0.7	-2.9		
	5%	DB	-5.3	-3.8	2.5	10.8	19.0	23.6	25.6	23.9	18.1	9.5	0.2	-4.3		
		MCWB	-5.8	-4.7	-0.3	5.0	10.2	13.6	15.8	15.5	12.1	6.4	-0.7	-5.0		
	10%	DB	-8.6	-6.6	-0.1	7.9	16.4	21.7	23.8	21.8	15.6	6.8	-1.4	-6.7		
		MCWB	-9.1	-7.3	-1.9	3.3	8.9	12.9	15.1	14.7	11.1	4.6	-2.2	-7.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	0.4	3.4	9.4	14.2	17.9	19.4	19.1	16.1	11.0	3.0	-0.7		
		MCDB	1.1	1.6	7.8	17.9	23.4	24.4	24.7	25.9	23.3	15.8	5.2	0.1		
	2%	WB	-3.0	-2.5	1.4	7.0	12.4	16.4	17.9	17.4	14.4	8.4	0.9	-2.9		
		MCDB	-2.5	-1.0	4.6	13.2	20.0	22.3	24.5	23.6	19.8	11.2	2.4	-2.1		
	5%	WB	-5.7	-5.1	0.0	5.2	10.9	15.2	17.0	16.4	12.9	6.5	-0.7	-4.9		
		MCDB	-5.1	-4.3	2.2	10.3	17.8	20.9	22.9	21.9	17.0	9.1	0.3	-4.2		
	10%	WB	-8.9	-7.7	-2.0	3.5	9.4	14.1	16.1	15.4	11.6	4.8	-2.1	-7.1		
		MCDB	-8.5	-7.1	0.0	7.5	15.3	19.6	21.7	20.2	14.9	6.8	-1.3	-6.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	13.4	15.0	13.4	13.1	12.4	12.0	10.9	9.6	7.2	8.2	9.9		
		MCDBR	11.4	13.2	14.7	15.2	17.8	15.9	15.6	14.8	14.0	11.4	8.2	9.5		
		MCWBR	10.9	12.6	11.1	9.2	9.1	7.1	6.7	7.0	7.5	7.4	6.9	8.9		
	5% WB	MCDBR	11.4	13.7	13.7	14.3	15.2	12.6	12.8	12.6	12.1	10.7	7.6	9.4		
		MCWBR	10.8	12.4	10.6	8.7	8.3	6.2	6.0	6.4	7.0	7.2	6.5	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.221	0.256	0.257	0.314	0.322	0.358	0.382	0.349	0.300	0.279	0.238	0.202		
	taud		2.230	2.241	2.315	2.295	2.486	2.409	2.367	2.459	2.613	2.573	2.321	2.173		
	Ebn at Noon		693	808	916	901	917	882	849	858	867	796	665	607		
	Edh at Noon		47	74	90	110	98	107	110	94	69	55	43	36		
(44) All-Sky Solar Radiation	RadAvg		0.69	1.73	3.40	4.99	5.66	5.74	5.30	4.27	2.82	1.41	0.59	0.41		
	RadStd		0.06	0.10	0.16	0.32	0.58	0.49	0.30	0.37	0.36	0.20	0.07	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.59	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	-0.30	-0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page